Spring 2026 Check-in

Segmentation Model Survey — JABr Condensates

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- Winter pipeline left a PC gap: automated 2.04 vs. manual reference 6.32
- Goal this week: survey alternative segmentation models to close the gap
- Dataset: JABr ROI Z-stack (55 slices × 185 × 259, 2-channel fluorescence)

Model Comparison

Same PC formula, same dataset — 4 new models benchmarked

Model	Approach	PC	Runtime
Old Cellpose (2D)	cyto2, slice-by-slice (baseline)	2.04	—
Cellpose 3 ★	**cyto3 + denoising + 3D mode**	**3.33**	**11 s**
StarDist 2D	versatile_fluo, slice-by-slice	1.91	6 s
U-FISH †	ONNX spot detection	3.81	2 s
Nellie	Frangi filter + graph analysis	3.57	14 s
Reference	**Manual / ImageJ benchmark**	**6.32**	**—**

★ Recommended † U-FISH uses Otsu threshold for nuclei (not a trained model) — PC likely inflated

Why Cellpose 3?

Three concrete changes over the Winter baseline (PC: 2.04 → 3.33)

1 Denoising

DenoiseModel cyto3 runs on every slice before segmentation — suppresses shot noise and sharpens condensate boundaries. PC intensities are always computed from the original raw pixels; denoising only guides boundary placement.

2 3D mode

do_3D=True processes XY, XZ, and YZ planes simultaneously and merges gradient flows before drawing boundaries. The old pipeline ran 55 independent 2D slices with no cross-slice awareness — a condensate spanning 3 slices was segmented 3 separate times, often with inconsistent boundaries, directly hurting the intensity ratio.

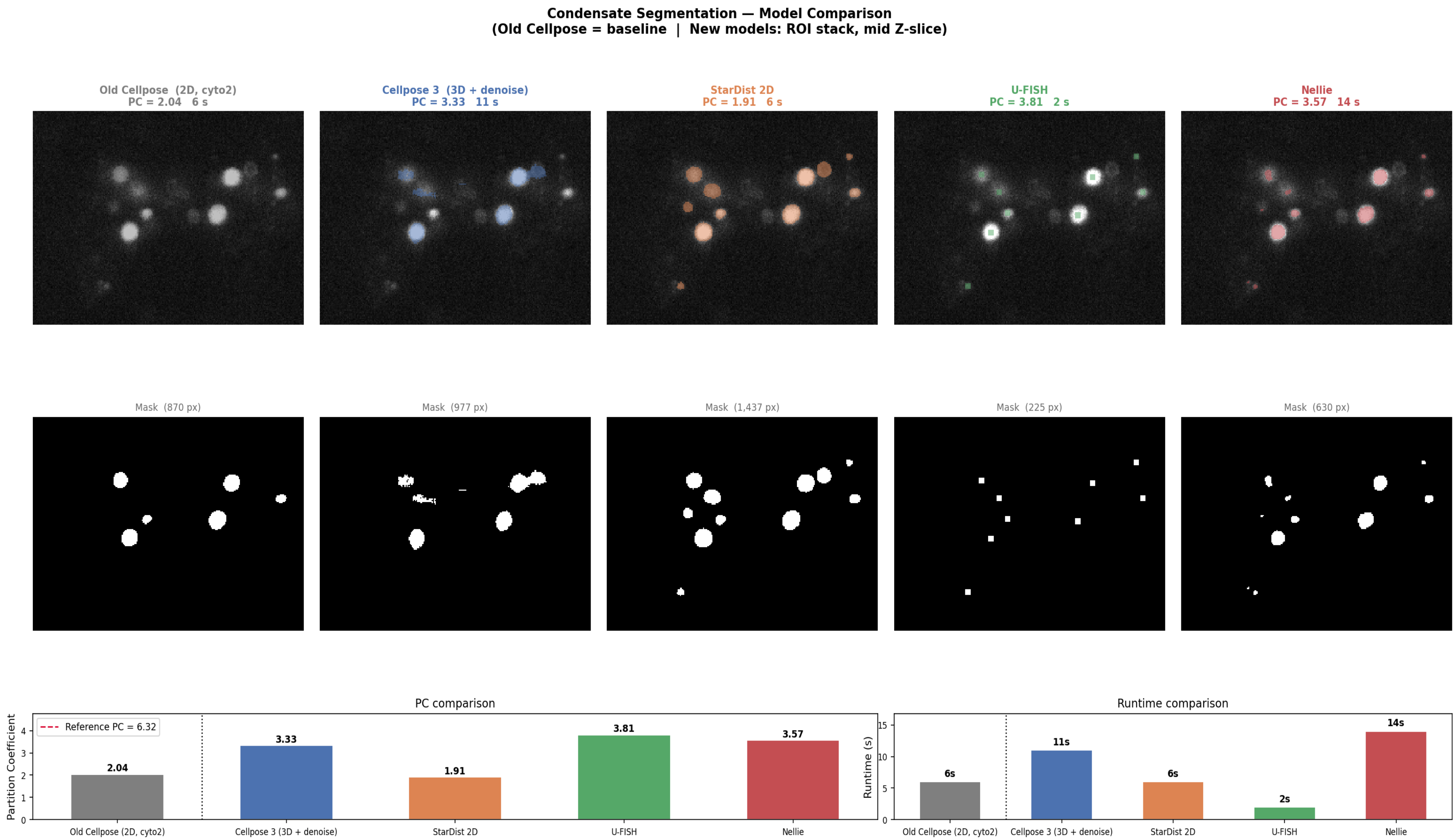
3 cyto3 vs cyto2

cyto3 is trained on a larger, more diverse dataset and is generally better at small, dim, or irregularly shaped objects. The 3D mode is likely the single largest contributor to the PC improvement.

> All models still fall short of reference PC = 6.32. The gap is due to condensate segmentation sensitivity — small boundary errors have an outsized effect on intensity ratios.

Segmentation Output — Side-by-Side

Mid-Z slice, all models



Caveats — Apples-to-Apples Problem

PC values are not fully comparable across models

Model	Condensate mask	Nuclei mask
Old Cellpose	Cellpose cyto2 (2D)	Cellpose cyto2 (2D)
Cellpose 3	Cellpose cyto3 (3D)	Cellpose cyto3 (3D)
StarDist 2D	StarDist2D	StarDist2D
U-FISH ■	**U-FISH spot + disk**	**Otsu threshold ← different!**
Nellie	Frangi + graph	Frangi + graph

Fix: use the same Cellpose 3 nuclear mask for all models and only swap the condensate mask — isolates condensate segmentation as the single variable.

Next Steps

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1 Standardize nuclear mask

Re-run all models using Cellpose 3 nuclei masks → fair PC comparison

2 3D volume estimation

Aggregate masks across Z-slices; compute true voxel volumes per condensate

3 CLI packaging

Replace hardcoded paths in cellpose_pipeline.py with argparse / config file; add docstrings

4 Spring poster / report

Narrative: problem → Winter pipeline → model survey → Cellpose 3 → 3D volume

Reference PC Investigation

2026-04-23 — found a methodological mismatch

Three types of PC

Nuclear PC — Condensate vs. dilute, both measured inside the nucleus. This is what the automated pipeline computes.

Cytoplasmic PC — Condensate vs. dilute, both measured in the cytoplasm. Not currently computed by the pipeline.

Background-subtracted PC — Same formula but a background constant B (measured outside all cells) is subtracted from both densities before taking the ratio. The reference CSVs use this method.

Findings for Sample2_5_1

Dilute density anomaly	Nuclear dilute density for Sample2_5_1 = 104.09, vs. ~70 for all other Sample2_5 entries. If dilute density were ~70, reference PC ≈ 9.4, not 6.32. Possible cause: loose manual ImageJ mask left bright condensate-edge pixels counted as dilute-phase background, deflating the reference PC.

Background subtraction not in pipeline	The person who computed the reference CSVs confirmed she applied background subtraction before computing PC. The automated pipeline does not. The formula becomes: PC = (condensate_density – B) / (dilute_density – B). Since dilute density is much closer to background than condensate density, subtracting B lowers the denominator more proportionally, inflating PC. Part of the gap (pipeline 3.33 vs. reference 6.32) is methodological, not just segmentation quality.

Action: confirm background estimation method with lab member → add background subtraction to pipeline → recompute PC for fair comparison with reference.